

Column 1 Sensitivity Analysis Report

Engineering Sensitivity Analysis of Boilup Ratio on Extractive Distillation Performance

1. Objective

The primary objective of this study was to investigate the influence of the **Column 1 boilup ratio** on the separation performance of the extractive distillation process.

During reproduction of the published literature model, several operating parameters required for complete simulation were unavailable. Consequently, a systematic engineering sensitivity analysis was performed to identify a stable operating region capable of achieving satisfactory ethanol purification while maintaining successful process convergence.

The study also served to determine whether the boilup ratio should be considered an important optimization variable during subsequent AI-assisted process optimization.

2. Base Case

The sensitivity analysis was carried out using the developed ChemSep process model while maintaining all operating conditions constant except the boilup ratio.

Parameter	Value
Feed Flow Rate	5173 kg/hr
Feed Composition	73 wt% Ethanol / 27 wt% Water
Solvent	Ethylene Glycol
Solvent Flow Rate	20,000 kg/hr
Solvent Temperature	256°C
Column Stages	15
Feed Stage	8
Solvent Stage	8
Column Pressure	3.5 bar

3. Variable Investigated

Only one operating parameter was varied.

Variable	Range
Column 1 Boilup Ratio	0.20 – 2.00

All remaining process conditions were maintained constant throughout the analysis.

4. Performance Indicators

The following performance indicators were monitored for each simulation.

- Ethanol Recovery
 - Total Energy Consumption
 - Reboiler Duty
 - Process Convergence
 - Temperature Profile
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5. Methodology

Starting from the validated base case, the boilup ratio was systematically varied over the selected operating range.

For every operating condition:

- the simulation was allowed to converge,
- process outputs were recorded,
- unrealistic operating conditions were rejected,
- energy requirements were calculated,
- engineering observations were documented.

The resulting data were then analysed to identify feasible operating regions for subsequent optimization studies.

6. Results

Figure 1- Effect of Boilup Ratio on Ethanol Recovery

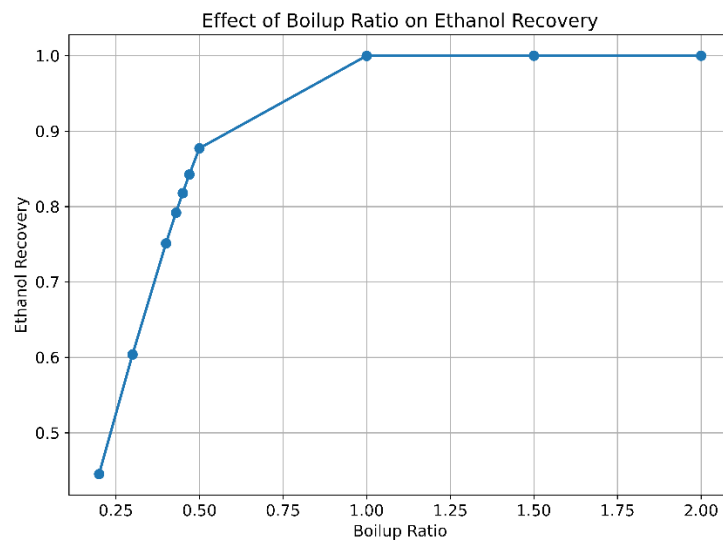


Figure 2- Effect of Boilup Ratio on Total Energy Consumption

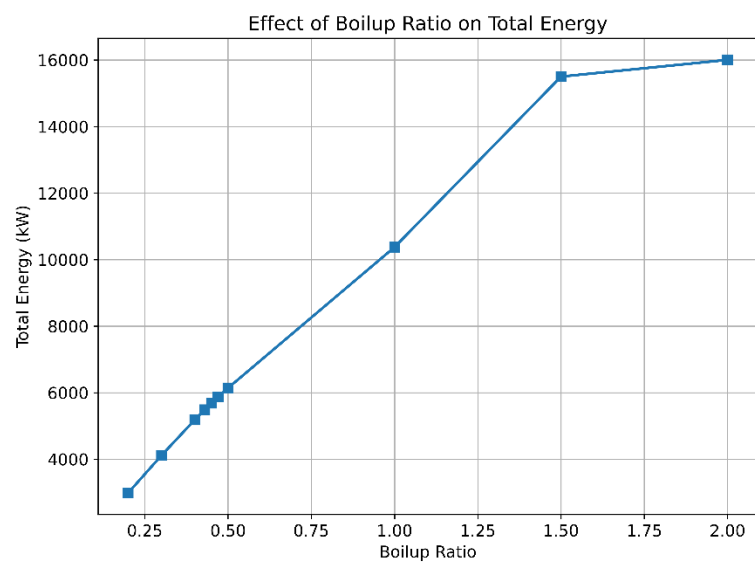


Figure 3 - Effect of Boilup Ratio on Reboiler Duty

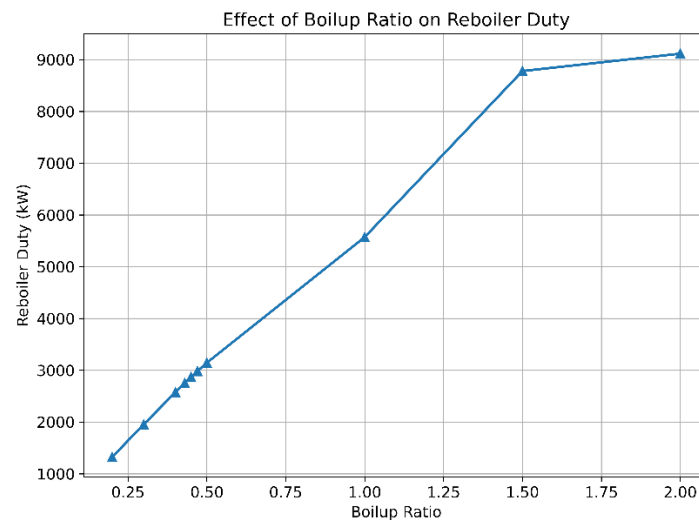
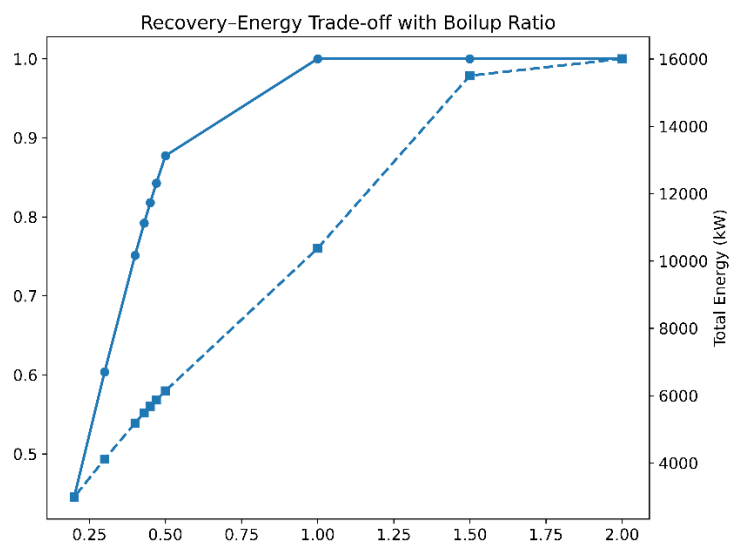


Figure 4- Recovery–Energy Trade-off



7. Engineering Discussion

The sensitivity analysis demonstrates that the boilup ratio is one of the most influential operating variables governing the performance of the extractive distillation column.

A direct relationship was observed between boilup ratio and ethanol recovery. Increasing the boilup ratio increased the internal vapor flow within the column, thereby enhancing mass transfer between the vapor and liquid phases. As a result, ethanol recovery increased continuously over the investigated operating range.

However, this improvement in separation performance was accompanied by a significant increase in energy consumption. Both the total energy requirement and the reboiler duty increased almost proportionally with increasing boilup ratio. Since additional vapor generation is achieved by supplying more heat to the reboiler, higher boilup ratios inevitably require greater thermal energy.

Although the highest boilup ratios produced nearly complete ethanol recovery, they also resulted in the largest operating energy requirements. Conversely, very low boilup ratios reduced energy consumption considerably but produced poor separation performance due to insufficient vapor generation inside the column.

This behaviour highlights the classical engineering trade-off between **product recovery** and **energy efficiency**. Therefore, selecting the maximum boilup ratio solely to maximize recovery would not represent the most economical operating condition.

Instead, an intermediate operating region was identified that provided stable convergence while maintaining acceptable ethanol recovery with substantially lower energy consumption.

The results clearly indicate that the boilup ratio should be considered one of the primary optimization variables in the overall process.

8. Engineering Findings

The following conclusions were obtained from the sensitivity analysis.

- Increasing boilup ratio increases ethanol recovery.
- Increasing boilup ratio increases total energy consumption.
- Increasing boilup ratio increases reboiler duty.
- The relationships observed were approximately monotonic over the investigated operating range.
- Boilup ratio exhibits the highest influence on Column 1 separation performance among the investigated variables.
- An engineering trade-off exists between recovery and energy consumption.
- A stable operating region was identified for use in subsequent optimization studies.

9. Conclusion

The sensitivity analysis confirmed that the boilup ratio is the dominant operating variable affecting the performance of the extractive distillation column. Increasing the boilup ratio improved ethanol recovery by enhancing vapor generation and mass transfer; however, this improvement required substantially higher thermal energy, resulting in increased reboiler duty and overall energy consumption.

Based on these observations, the boilup ratio was selected as one of the principal optimization variables for subsequent dataset generation and machine learning-assisted process optimization. The feasible operating range established in this study was later used to generate the engineering dataset employed throughout the remainder of the project.